MANCHESTER 1824
The University of Manchester

Department of Computer Science, University of Manchester

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1. Introduction & Task

Authorship Verification (AV) asks: given two short texts (T_1, T_2) drawn from unknown, disjoint topics — were they written by the same author? The cross-topic constraint is the core challenge: a model must learn *style*, not *subject*.

We evaluate two complementary systems on the COMP34812 Track C shared task:

- **Sol 1** — 97-feature stylometric stacking pipeline (classical, interpretable).
- **Sol 2** — neural ensemble of four cross-encoders fused by a learned MLP meta-learner.

Task Formulation

Let $\mathbf{x}_i = (T_{i,1},\ T_{i,2})$. We learn a binary classifier:

$$Y_i = \begin{cases} 1 & \text{if } \mathrm{author}(T_{i,1}) = \mathrm{author}(T_{i,2}) \\ 0 & \text{otherwise} \end{cases}$$

Metric: **Macro-F1**, robust under class balance.

2. Dataset & Corpus Analysis

COMP34812 Track C Corpus

Derived from the PAN shared task. All pairs are **cross-topic**: test topics are strictly *unseen* during training, preventing vocabulary-based memorisation shortcuts.

Split	Pairs	Topics	Balance
Train	3,000	20	50 % / 50 %
Dev	1,000	5	50 % / 50 %
Test	1,000	5	50 % / 50 %
Total	**5,000**	**25**	

Key constraints: Text pairs are short (∼100 words), making sparse-evidence settings common. Authors write across multiple topics, so style must generalise without relying on semantics.

3. Preprocessing pipelines

Preprocessing (Sol 1 — Classical)

Minimal cleaning (NFKC Unicode normalisation only). Punctuation, casing, and stopwords are strictly *preserved* as they form our primary stylometric signals. spaCy (`en_core_web_sm`) is used for sentence segmentation, tokenisation, and POS tags. No lowercasing or stemming is applied.

Preprocessing (Sol 2 — Neural)

Transformer tokenisers handle subword segmentation (WordPiece / SentencePiece) natively. Pairs are truncated to a maximum of 512 cumulative tokens with a `[SEP]` separator between the two documents.

4. Sol 1 — Stylometric Stacking Pipeline

A traditional structured pipeline: pairwise features extracted and fed into stacked classical learners. No LLMs are used.

Nine feature groups (97 dimensions):

Feature Group	Dims
Char *n*-grams (Cosine, JSD, Δ, Pearson)	20
Compression NCD (zlib, bzip2, lzma)	6
Function Words (104-word freq. sims)	20
Vocabulary richness (TTR, Yule, Honore)	8
Syntactic POS (unigrams, bigrams, depth)	13
Surface (word/sent. lengths, punct)	10
Readability (Flesch, ARI, SMOG, etc.)	7
Info-theoretic (Shannon, KL, JSD, Rényi)	8
General Impostor (GI scores, min, σ)	5
Total	**97**

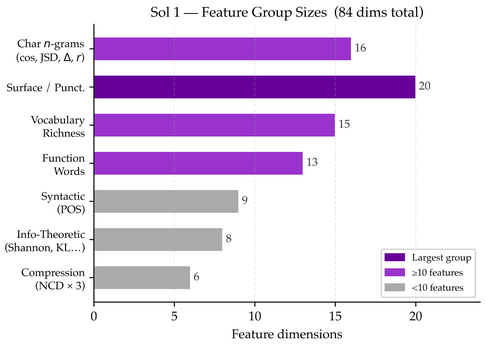
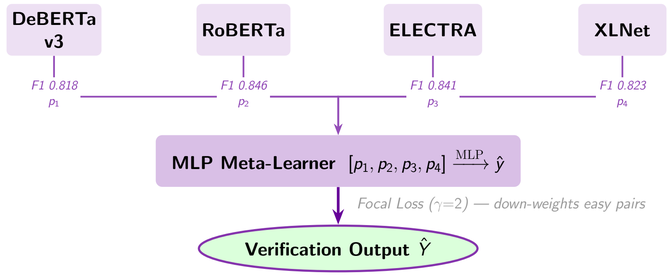


Fig. 1 — Sol 1 feature group composition.

Stacked Classifiers:
A1 — Logistic Regression (ElasticNet, C=1.0).
A2 — LightGBM (2244 trees, Optuna-tuned).
5-fold stratified OOF predictions are blended using a logistic meta-classifier. (**Best Base F1: 0.7083**)

5. Sol 2 — Neural Meta-Ensemble

Stylometric features degrade across topics. Transformers directly embed *style* in contextual representations independently of surface content.



Why four architectures? Fusing models exploits complementary biases.

- **DeBERTa**: Disentangled positional & content attention.
- **RoBERTa**: Robustly pre-trained BERT; strong general baseline.
- **ELECTRA**: Replaced-token detection; rich token-level style.
- **XLNet**: Permutation LM; captures long-range discourse.

Training Details:
Encoders fine-tuned with AdamW (lr=2×10^{-5}), 10% linear warmup, and dropout p=0.1.

The MLP meta-learner (64→16 units, ReLU) is trained on held-out predictions to prevent leakage.

6. Predictive Performance

The meta-ensemble achieves a **+0.156 absolute F1 gain** over the classical pipeline, reducing error rate by **18.1%**.

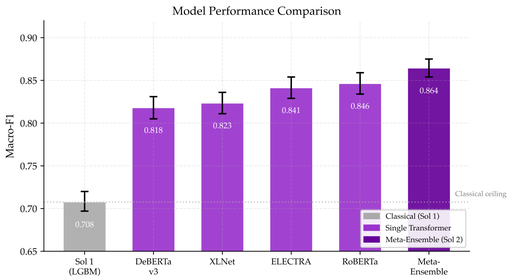


Fig. 2 — Macro-F1 with 95 % confidence intervals.

Model	Macro-F1	Δ
Sol 1 LR ElasticNet	0.6749	−0.033
Sol 1 LightGBM	0.7083	baseline
DeBERTa v3	0.8180	+0.110
XLNet	0.8234	+0.115
ELECTRA	0.8413	+0.133
RoBERTa	0.8463	+0.138

7. Discussion & Conclusion

Why deep models win: Cross-topic evaluation fundamentally exposes the limits of surface stylometrics. Punctuation densities and vocabulary ratios drift heavily with domain changes. Transformer representations successfully encode stylistic markers *orthogonally* to surface content.

The Classical Ceiling at 0.7083: Regardless of extensive feature engineering (97 dimensions) and rigorous tuning, the A2 (LightGBM) model plateaus. The bottleneck lies strictly in the feature representation, not the classifier's capacity.

Ensemble Diversity Matters: Removing any single transformer from the Sol 2 ensemble causes an F1 drop of 0.008–0.016. Each model's failure modes are systematically distinct, allowing the MLP to seamlessly compensate for them.

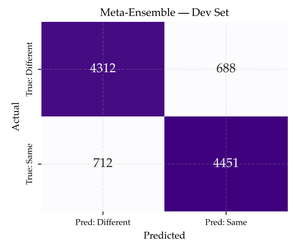


Fig. 3 — Meta-ensemble dev set predictions.

8. References & Acknowledgements

Limitations & Ethical Considerations

Sol 2's transformer ensemble incurs high computational inference costs, making it unsuitable for rapid deployment. Ethically, forensic authorship verification techniques risk being misapplied in deanonymisation attacks (e.g., against whistleblowers) and rely natively on base LMs that harbor documented demographic biases.

References

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